

ASSIGNMENT - 07  
NAME - PRERANA MUDIAR  
ROLL NO - CSB24021

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from typing import List, Dict, Set
from collections import defaultdict
from functools import reduce

# Total screen time for each user
def total_time_per_user(logs: List[dict]) -> Dict[str, float]:
    user_time = defaultdict(float)

    for log in logs:
        user_time[log["user"]] += log["duration"]

    return dict(user_time)

# Top K most active users
def most_active_users(logs: List[dict], k: int) -> List[str]:
    totals = total_time_per_user(logs)

    return [user for user, _ in
            sorted(totals.items(), key=lambda x: x[1], reverse=True)[:k]]

# All unique actions performed
def unique_actions(logs: List[dict]) -> Set[str]:
    return {log["action"] for log in logs}

# Total activity time using reduce()
def total_activity_time(logs: List[dict]) -> float:
    return reduce(lambda acc, log: acc + log["duration"], logs, 0.0)

# Example data
logs = [
    {"user": "CSB24021", "action": "YouTube", "duration": 1.5},
    {"user": "CSB24042", "action": "Instagram", "duration": 2.0},
    {"user": "CSB24021", "action": "WhatsApp", "duration": 0.5},
    {"user": "CSB24043", "action": "YouTube", "duration": 3.0}
]

print(total_time_per_user(logs))
print(most_active_users(logs, 2))
print(unique_actions(logs))
print(total_activity_time(logs))
```

```
print(total_time_per_user(logs))
print(most_active_users(logs, 2))
print(unique_actions(logs))
print(total_activity_time(logs))
```

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```
{'CSB24021': 2.0, 'CSB24042': 2.0, 'CSB24043': 3.0}
['CSB24043', 'CSB24021']
{'YouTube', 'Instagram', 'WhatsApp'}
7.0
```

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